

Document Title: XR-500 Industrial Cooling Controller Installation Manual — Revised
Specification Product Name: XR-500 Industrial Cooling Controller Document Type: Installation
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1. **Product Overview** The XR-500 Industrial Cooling Controller is an advanced digital control unit designed for precise thermal management of industrial chillers, heat exchangers, and secondary fluid cooling systems. Built around a high-performance 32-bit microcontroller core, the unit regulates compressor staging, pump speeds, and motorized valve actuators. The chassis is housed in a heavy-duty IP65 die-cast aluminum enclosure designed for harsh manufacturing environments. This revised manual outlines updated specifications, mechanical installation guidelines, and electrical hookup requirements for Rev 3.0 hardware configurations.
2. **Technical Specifications**
 - Power Supply Input: 24V DC $\pm 10\%$ (Regulated Class 2 supply)
 - Current / Power Consumption: 2.0 A max / 48 W
 - Sensor Input Channels: 4x PT1000 RTD (3-wire configuration)
 - Analog Outputs: 4x 0–10V / 4–20 mA selectable
 - Digital Relay Outputs: 4x SPDT, rated 250V AC, 5A (resistive)
 - Maximum System Operating Pressure: 3.5 MPa (35 bar)
 - Coolant Circulation Flow Rate: 50 L/min minimum to 200 L/min maximum (Nominal: 130 L/min)
 - Network Connectivity: Modbus RTU (RS-485), BACnet/IP, Modbus TCP
 - Operating Ambient Temperature: -10°C to 55°C
 - Protection Class: IP65
 - Physical Dimensions: 240 mm x 180 mm x 90 mm
 - Net Weight: 3.5 kg
3. **Installation Requirements** Mount the XR-500 vertically on a rigid structural bulkhead or mounting plate rated for at least 20 kg static load. Fasten using four M6 corrosion-resistant machine screws torqued to 4.5 N·m. Maintain a minimum perimeter clearance of 150 mm above, below, and beside the enclosure to support adequate thermal dissipation via natural convection. Electrical cabling must pass through the bottom cable entry plate using IP65-rated liquid-tight glands. Torque cable glands to 3.5 N·m upon installation. Separate signal lines (sensors and communications) from mains/relay power wiring by placing them in distinct conduit runs spaced at least 100 mm apart. RS-485 serial communication cabling must utilize 120-ohm shielded twisted pair, with the drain wire terminated directly to the chassis ground lug at the controller end.
4. **Operating Conditions** The unit is engineered for continuous industrial operation in ambient environments ranging from -10°C to 55°C , with non-condensing relative humidity levels between 5% and 95%. Operation at altitudes up to 2000 meters requires no derating; above 2000 meters, derate maximum ambient operating temperature by 0.5°C per 100 meters elevation gain. Avoid placing the controller in direct sunlight to prevent thermal saturation of the enclosure. The fluid circuit must use water/glycol

mixtures (up to 50% glycol) or treated water. The system hydraulic circuit must be maintained at a working pressure below 3.5 MPa. Operating below the minimum flow threshold of 50 L/min will generate a low-flow warning and halt staging sequences to protect connected equipment.

5. Safety Information Installation, configuration, and electrical servicing must be carried out by trained industrial automation technicians. High AC voltage (up to 250V AC) may be connected to relay contacts even when the main 24V DC power feed to the controller is disconnected. Disconnect and lock out all external electrical circuits before opening the enclosure cover. Standard electrostatic discharge (ESD) precautionary measures must be observed when handling internal electronics. Do not subject the power input terminals to voltages exceeding 30V DC to prevent permanent internal component failure.
6. Maintenance / Service Procedures Perform a preliminary hardware check before energizing the system. Inspect all electrical connections and ensure screw terminal blocks are tightened to a torque of 0.5 N·m. Verify that enclosure sealing gaskets are properly seated and undamaged. On an annual basis, verify calibration of PT1000 sensor inputs using an approved calibration standard. Check the integrity of cable glands every six months in environments subject to vibration.
7. Troubleshooting
 - Symptom: Controller does not power up. Possible Cause: Main 24V DC power source disconnected, low voltage, or reverse polarity wiring. Action: Verify supply output voltage at terminals is within 21.6V DC to 26.4V DC. Check wire orientation.
 - Symptom: Error Code E-102 (System High Pressure). Possible Cause: Hydraulic pressure exceeding max specification. Action: Inspect fluid lines and ensure line pressure is kept below the 3.5 MPa limit. Check pump performance and valve positioners.
 - Symptom: Error Code E-204 (Insufficient Flow Rate). Possible Cause: Circulation flow rate dropping below 50 L/min. Action: Inspect inline filters, clear obstructions, and check pump speeds to restore normal 130 L/min operation.
 - Symptom: Serial communication fault on RS-485 network. Possible Cause: Termination resistor mismatch or improper ground shield termination. Action: Enable 120-ohm termination resistor only at the physical end nodes of the bus. Re-check single-point shield connection.
8. Additional Notes System updates are loaded via the primary RJ45 Ethernet port using the XR-500 Service & Configuration Tool software (v3.2 or later). Always save system setup parameters prior to executing firmware flashes. The integrated real-time clock (RTC) is backed up by a CR2032 coin cell battery with an anticipated operational life of 5 years. Dispose of used batteries according to local hazardous waste guidelines.
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